ATHIYAMAN M



Software Engineer | Backend Systems | Applied AI & LLMs | Healthcare Tech

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Portfolio: https://athiyaman-m.github.io/

PROFESSIONAL SUMMARY

Software Engineer with experience in Python backend development, security monitoring systems, database-driven applications, and AI-powered solutions. Currently working on backend engineering and examination security infrastructure at the Centre for Development of Advanced Computing (**C-DAC**), Government of India. Previously contributed to clinical software development and medical-image processing at the Healthcare Technology Innovation Centre (**HTIC**), IIT Madras.

Experienced in developing REST APIs, risk-scoring pipelines, data-processing workflows, and LLM/RAG-based applications. Research and development interests include explainable AI, clinical decision-support systems, biomedical data analysis, and intelligent software systems. Combines practical engineering experience with academic research, interdisciplinary project development, and technical problem-solving.

RESEARCH & TECHNICAL INTERESTS

- Applied Artificial Intelligence, Machine Learning, and Deep Learning.
- Explainable AI (XAI), model interpretability, and transparent decision-support systems.
- Large Language Models (LLMs), RAG, AI agents, and multi-model orchestration.
- Clinical AI, biomedical data analysis, computational oncology, survival modelling, and clinical digital twins.
- Medical-image processing, 3D anatomical visualization, and healthcare technology.
- Backend engineering, PostgreSQL, database optimization, security monitoring, anomaly detection, and scientific-document intelligence.

PROFESSIONAL EXPERIENCE

Project Engineer - Backend Software Engineer

Sep 2025 - Present | Chennai, India

Centre for Development of Advanced Computing (C-DAC), Government of India

Contributed to backend software development, security analytics, database, and AI proof-of-concept development within government initiatives.

- Architected and shipped **FastAPI backend services** powering a **real-time security monitoring and risk-scoring system** deployed on PXE servers, enabling live monitoring of **thousands of client systems** during national SSC examinations.
- Developed **Python-based scoring workflows** using FastAPI, SQL, SQLite and MariaDB to process client-security indicators, calculate machine-level risk scores, and update examination-security records.
- Implemented **severity-based scoring for security flags** involving network ports, firewall events, unauthorized activities, machine integrity, and client status during national SSC examinations.
- Integrated **FastAPI and REST APIs** to support security-data exchange, database operations, machine-level and examination-centre-level analytics.
- Investigated scoring discrepancies and false positives using structured **logging**, regex-based log parsing, SQL queries, and source-code analysis, contributed to centralized monitoring using **Elasticsearch and Kibana**.
- Worked on dynamic scoring configuration by exploring and implemented **cryptography algorithms**, encryption/decryption, hashing configuration distribution, API-driven updates, database-backed alternatives and integrity verification.
- Built anomaly detection, device validation, and automated risk-assessment pipelines using **30+ security** indicators across **5+ data source**s, with severity-based scoring and API-driven blocking for real-time enforcement.
- Built Support **PostgreSQL** from source on Linux and analysed database internals, including aclchk.c, concurrent GRANT/REVOKE locking behaviour, and B-tree skip-scan optimization.
- Conducted concurrency testing and pgbench benchmarking, recording an approximately **8% performance improvement** in one patch iteration, investigated and **resolved three open-source PostgreSQL bugs**.
- Developed **AI proof-of-concepts** for financial document intelligence, research summarization, and explainable healthcare applications using **LLMs and RAG**, prepared technical architecture and patent-oriented documentation.

Project Intern - Healthcare Software Developer

Jun 2024 - Jul 2025 | Chennai, India

Healthcare Technology Innovation Centre (HTIC), IIT Madras

Contributed to clinical software development and medical-image processing for Total Knee Arthroplasty surgical planning.

- Developed and enhanced clinical software workflows supporting **TKA planning using Qt and C++**.
- Worked with **VTK** and **ITK** libraries for **medical-image processing**, anatomical data handling, segmentation workflows, and **3D visualization**.
- Contributed to **CT-based bone segmentation and reconstruction** workflows that transform medical-imaging data into visual representations for surgical-planning applications.
- Integrated **REST APIs and MySQL**-based data management into healthcare application workflows.
- Used **Git, Azure DevOps, and CMake** for code management, software builds, collaborative development, and testing.
- Worked in an interdisciplinary clinical-technology environment where software functionality needed to align with medical-image data, anatomical geometry, and practical surgical-planning requirements.
- Gained experience translating clinical workflow requirements into software modules and image-processing operations.

PROJECTS

1. ExAI-PanC: Explainable AI Clinical Digital Twin for Pancreatic Cancer

Objective: Develop an integrated AI research prototype combining multi-omics feature learning, pancreatic cancer classification, survival prediction, explainability, and interactive 3D clinical visualization.

Work / Implementation:

- Developed a multi-omics analysis pipeline using 183 TCGA-PAAD cases and 362 GTEx samples, incorporating VAE-based feature learning and XGBoost classification.
- Integrated Random Survival Forest modelling and SHAP-based explanations to analyse survival outcomes and interpret model predictions.
- Built an interactive 3D clinical digital twin featuring an 11-organ risk heatmap, biomarker visualization, ODE-based disease-progression simulation, and comparison of four chemotherapy regimens.

Contribution: Integrated predictive modelling, survival analysis, explainable AI, and 3D anatomical visualization into a unified pancreatic cancer research platform. **Research Limitation:** Reported model performance requires careful validation for data leakage, generalizability, and clinical applicability; the prototype is not clinically validated.

Tech Stack: Python, PyTorch, XGBoost, SHAP, scikit-learn, scikit-survival, Streamlit, Plotly, pandas, NumPy

Repository: https://github.com/athiyaman-m/XAI-PanC

2. AuditShakti - AI Compliance & Audit Intelligence Platform

Objective: Develop an AI-powered document-intelligence platform for financial document analysis, regulatory compliance checking, and evidence-based audit reporting.

Work / Implementation:

- Built a document-processing pipeline using PDF extraction and OCR to analyse financial documents and support compliance checks across Ind-AS, SEBI, RBI, and ESG-related requirements.
- Implemented hybrid retrieval combining SQLite FTS5 and semantic search to retrieve relevant evidence and support document-grounded queries.
- Developed a FastAPI backend and Next.js interface with conversational AI, page-level evidence traceability, and PDF, Excel, and JSON report exports.

Contribution: Integrated document extraction, configurable compliance validation, hybrid retrieval, explainable findings, and automated reporting into a unified audit-intelligence prototype. **Research Limitation:** Regulatory accuracy and reductions in manual audit effort require systematic evaluation against expert-validated documents and compliance outcomes.

Tech Stack: Python, FastAPI, Next.js, React, TypeScript, scikit-learn, PyMuPDF, Tesseract OCR, Gemini API, SQLite, FTS5, semantic search

Repository: https://github.com/athiyaman-m/AuditShakti-README

3. Research Paper Intelligence - Multi-LLM RAG System

Objective: Develop a research assistant for scientific-paper summarization, cross-paper comparison, and evidence-oriented research synthesis using multiple LLMs and RAG techniques.

Work / Implementation:

- Built a section-aware PDF-processing pipeline using PyMuPDF to extract research content and organize papers for structured summarization.
- Integrated hosted and locally executed language models through LangChain, Groq API, and Ollama to support summarization and comparative analysis.
- Implemented evaluation workflows for ROUGE-2 F1, factual consistency, and section coverage, reporting improvements of 98%, 55%, and 33%, respectively, against the documented baseline.

Contribution: Developed a multi-LLM research workflow combining structured document extraction, model orchestration, comparative synthesis, and quantitative evaluation.. **Research Limitation:** Reported improvements are specific to the evaluated dataset and baseline; broader performance requires reproducible testing across diverse scientific papers and models.

Tech Stack: Python, Streamlit, LangChain, Groq API, Ollama, Llama 3.3, Gemma, PyMuPDF

Repository: https://github.com/athiyaman-m/Research-Paper-Summarizer

4. Explainable Mental Health Risk Prediction System

Objective: Develop an interpretable machine-learning prototype for mental-health risk classification using contextual text representations and structured behavioural indicators.

Work / Implementation:

- Combined multilingual BERT text embeddings with sentiment features, stress-related keywords, and structured indicators such as sleep duration, attendance, screen time, and academic-performance changes.
- Implemented a classification pipeline for low-, medium-, and high-risk categories using scikit-learn, with SHAP and LIME for feature attribution and individual prediction explanations.
- Developed a Flask API and Streamlit dashboard with SQLite-backed feedback collection, counterfactual exploration, and prediction-performance monitoring.

Contribution: Integrated contextual NLP, structured-data classification, explainability, and human-in-the-loop feedback into an interpretable AI research prototype. **Research Limitation:** Reported performance requires independent validation, bias assessment, and expert evaluation. The system is not a clinically validated diagnostic tool.

Tech Stack: Python, PyTorch, Hugging Face Transformers, multilingual BERT, scikit-learn, SHAP, LIME, Flask, Streamlit, Plotly, SQLite

Repository: https://github.com/athiyaman-m/mhp-xai

PUBLICATIONS

1. Mental Health Risk Detection via Explainable AI (XAI): A Multi-Model Benchmark with Contextual Embeddings

- **Status:** Accepted, IEEE ICAITPR 2026
- **Research area:** Explainable AI, contextual text representations, mental-health risk classification, and comparative model evaluation.
- **Research focus:** Investigates a multi-model evaluation framework emphasizing contextual embeddings and model interpretability.

2. AI-Based Analysis of Hospital Bed Utilization Patterns for Patient Readmission Prediction

- **Status:** Accepted, SAHITHYA 2025
- **Research area:** Healthcare analytics, hospital utilization, patient readmission prediction, and machine learning.
- **Research focus:** Examines predictive modelling approaches for hospital-bed utilization and readmission-related analysis.

3. Design of PON for Hospital Building Management: Enhancing Data Transmission Using FSO

- **Status:** Accepted, ICHCSC 2025, Springer Nature.
- **Research area:** Passive optical networks, free-space optical communication and hospital-building network infrastructure.
- **Research focus:** Investigates optical-network design and free-space optical communication for data transmission in hospital-building management.

EDUCATION

B.E. Electronics & Communication Engineering (Hons. Artificial Intelligence) *Nov 2021 - May 2025 | Tamil Nadu, India*

IFET College of Engineering, Anna University | CGPA: **8.71/10** | **Rank Holder** | **First Class with Distinction**

SKILLS

Programming: Python, SQL, Bash, C++
Backend & APIs: FastAPI, Flask, RESTful HTTP APIs (JSON payloads), Requests, Microservices, ETL Pipelines
Databases: PostgreSQL, MariaDB, MySQL, SQLite, Elasticsearch, Pinecone, MySQLdb, PyMySQL
AI / ML / GenAI: LLMs, RAG, LangChain, LangGraph, NLP, Hugging Face, OpenAI / Groq / Gemini API, PyTorch, scikit-learn, SHAP/LIME, Pandas, NumPy
DevOps & Tools: Git, GitHub, Docker, Linux, Azure DevOps, CI/CD, Kibana/ELK Stack, Power BI

ACHIEVEMENTS & AWARDS

- Awarded the **Rank Holder Award** and **Academic Excellence Award** for outstanding academic performance.
- Recognized as the **Best Student** for consistent academic excellence, leadership contributions, and overall achievements.
- Received the **Best Innovation Project Award** at Project Expo'25 for developing an impactful technology-based solution.
- Awarded **Best Presentation** in the Technical Talk Competition for excellence in technical and research presentation.
- Secured **Runner-Up** position in the CEMP Project Expo 2022 for innovative project development and implementation.
- **Finalist** in a National-Level Mathematics Competition, demonstrating strong analytical and problem-solving abilities.
- Earned the **LinkedIn Top Voice Badge** on three occasions for sharing technical knowledge and industry insights.
- Recognized by **C-DAC for outstanding work performance** and contributions to software development projects.

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Data Analytics Professional Certificate - **Google**
- Power BI Analyst - **Microsoft**
- Python for Data Science - **IBM**
- Generative AI (LinkedIn Learning) - **Microsoft**
- Prompt Engineering (Cognitive Class) - **IBM**
- Software Engineering Architecture - **Accenture**

LEADERSHIP AND PROFESSIONAL ENGAGEMENT

- **Class Representative** - undergraduate academic coordination and student communication.
- **Associate Secretary** - Electronics and Communication Engineering.
- **ISTE Chapter** - student and professional technical activities.
- **Google Developer Groups (GDG) Chennai** - professional community engagement.
- **Technical project evaluation / jury participation** - College students project review, discussion and supporting.

REFERENCES

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